

Activity: Genetic Algorithm

For Sections: B, E, F

Background

The **n-queens problem** involves placing n queens on an $n \times n$ chessboard such that no two queens attack each other. To adapt this to a Genetic Algorithm, we represent the board as a *chromosome*: an array where the index is the column and the value is the row position (1–8).

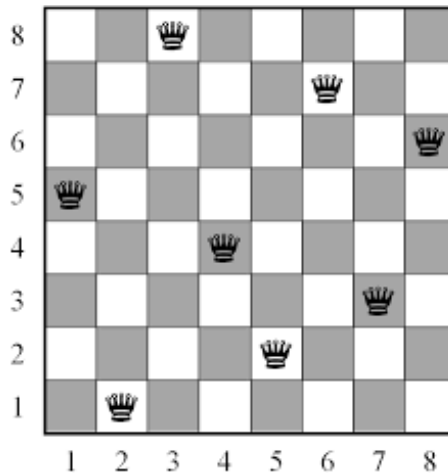


Figure 1: Example Chromosome: [5, 1, 8, 4, 2, 7, 3, 6]

Part 1 :

Consider the following initial population of four parents:

- **P1:** [2, 4, 1, 8, 3, 6, 7, 5]
- **P2:** [1, 5, 2, 6, 3, 7, 4, 8]
- **P3:** [4, 2, 7, 1, 8, 5, 3, 6]
- **P4:** [8, 6, 4, 2, 7, 5, 3, 1]

1.1 Fitness Evaluation

The fitness $f(x)$ is the number of **non-attacking pairs**. (Max fitness = 28).

$f(P1) = \underline{\hspace{2cm}}$ $f(P2) = \underline{\hspace{2cm}}$

$f(P3) = \underline{\hspace{2cm}}$ $f(P4) = \underline{\hspace{2cm}}$

1.2 Crossover

Perform **Single-Point Crossover** after the specified split points.

Constraint: This problem requires a *permutation* (each row 1–8 must appear exactly once). If an offspring has duplicate values, **repair it** by replacing all duplicates in the second half of the chromosome with the missing column numbers (in ascending order).

1. **Offspring A** (P1 and P2, split after index 3 (column 4)): [, , , , , , ,]
2. **Offspring B** (P2 and P1, split after index 3 (column 4)): [, , , , , , ,]
3. **Offspring C** (P3 and P4, split after index 5 (column 6)): [, , , , , , ,]
4. **Offspring D** (P4 and P3, split after index 5 (column 6)): [, , , , , , ,]

1.3 Mutation

Apply a random mutation to **Offspring A** and **Offspring C**:

- **Mutated A** (Change index 6 to value **2**): [, , , , , , ,]
- **Mutated C** (Change index 0 to value **1**): [, , , , , , ,]

Part 2: Conceptual Logic

1. If the entire population has a “1” at index 0 and the mutation rate is **0%**, can the algorithm ever place a queen in Row 5 for Column 1? Explain.
2. In Task 1.2, we used a **Repair Function** to maintain the constraint.
 - What would happen to the fitness score if we left the duplicates?
 - If there is no repair function, how does this affect the search space (does it make it larger or smaller)?